As a profitable example of the application of this law of synthesis, in its present extended form, let it be required to determine the conditions that a function of x, y of the fifth degree may have three equal roots. In general, let $\phi = ax^5 + 5bx^4y + 10cx^3y^2 + 10dx^2y^3 + 5exy^4 + fy^5$, then ϕ has a quadratic and cubic covariant of which I have written at large in my supplemental essay above referred to, being in fact the s and t (that is the quadrinvariant and cubinvariant) in respect to x', y' (x, y being treated as constants) of

$$\left(x'\frac{d}{dx} + y'\frac{d}{dy}\right)^4 \phi.$$

Let these covariants respectively be called

$$Ax^2 + 2Bxy + Cy^2 = u,$$
$$\alpha x^3 + 3\beta x^2y + 3\gamma xy^2 + \delta y^3 = v;$$

then

$$\left.\begin{matrix} Ax + By \\ Bx + Cy \end{matrix}\right\}$$

forms a plexus, and

$$\left.\begin{matrix} \alpha x^2 + 2\beta xy + \gamma y^2 \\ \beta x^2 + 2\gamma xy + \delta y^2 \end{matrix}\right\}$$

will form another.

Now when $a = 0$, $b = 0$, $c = 0$, ϕ will have three equal roots, and

$$\left(x'\frac{d}{dx} + y'\frac{d}{dy}\right)^4 \phi$$

becomes

$$6dy \,.\, x'^2y'^2 + 4(dx + ey)\,x'y'^3 + (ex + fy)\,y'^4,$$

of which the quadrinvariant in respect to x', y' is easily seen to be d^2y^2 and the cubinvariant d^3y^3. Accordingly the grouping

$$\begin{matrix} A, & B \\ B, & C \end{matrix} \text{ becomes } \left.\begin{matrix} 0, & 0 \\ 0, & d^2 \end{matrix}\right\},$$

and the grouping

$$\begin{matrix} \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{matrix} \text{ becomes } \left.\begin{matrix} 0, & 0, & 0 \\ 0, & 0, & d^3 \end{matrix}\right\}.$$

Accordingly, we see that the determinant $\begin{vmatrix} A, & B \\ B, & C \end{vmatrix}$ and all the first minors of $\begin{vmatrix} \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{vmatrix}$, that is $\alpha\gamma - \beta^2$, $\beta\delta - \gamma^2$, $\alpha\delta - \beta\gamma$, become zero; but the former single quantity $\begin{vmatrix} A, & B \\ B, & C \end{vmatrix}$ being an invariant, and this last system being an invariantive plexus, all the quantities so affirmed to be zero will remain zero, notwithstanding any linear transformations to which ϕ may be subjected; thus then we obtain an immediate proof of the theorem that when a function of x and y of the fifth degree contains three equal roots the determinant of its quadratic covariant, which in fact is its sole quart-invariant, and the first minors of its cubinvariant will be all separately zero. This theorem may be made still more stringent; for by combining

$$Ax^2 + 2Bxy + Cy^2,$$

$$\alpha x^2 + 2\beta xy + \gamma y^2,$$

$$\beta x^2 + 2\gamma xy + \delta y^2,$$

it becomes manifest that in the case supposed all the first minor determinants of

$$\begin{vmatrix} A, & B, & C \\ \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{vmatrix}$$

will be zero, showing in addition to the theorem last enunciated that also

$$A : B : C :: \alpha : \beta : \gamma :: \beta : \gamma : \delta.$$

It is curious and instructive to remark that this last set of equations, stringent as *they appear*, and far more than enough to express a duplex condition, are not sufficient to imply unequivocally the existence of three equal roots, unless we have also $AC - B^2 = 0$; for suppose ϕ to take the form $ax^5 + fy^5$ (b, c, d, e all vanishing); then it will easily be seen that

$$\alpha = 0, \quad \beta = 0, \quad \gamma = 0, \quad \delta = 0,$$

$$A = 0, \quad B = af, \quad C = 0.*$$

* If we take L, M, N a system of fundamental invariants to ϕ, of which all the other invariants of ϕ are rational integer functions, then $L = \begin{vmatrix} A, & B \\ B, & C \end{vmatrix}$ and the simplest forms for M and N are

$$M = \begin{vmatrix} A, & B, & C \\ \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{vmatrix} \quad \text{and} \quad N = \begin{vmatrix} \alpha, & 2\beta, & \gamma & \\ & \alpha, & 2\beta, & \gamma \\ \beta, & 2\gamma, & \delta, & \\ & \beta, & 2\gamma, & \delta \end{vmatrix},$$

where L and N are the discriminants of the quadratic and cubic covariants of ϕ respectively, and a linear function of M, L^2 is the discriminant of ϕ itself (L, M, N being of 4, 8, and 12 dimensions respectively in the coefficients of ϕ).

For many purposes of the calculus of forms it is desirable to have the command of cases for which any two out of these three invariants may be made to vanish without the third vanishing; and it will be found that when ϕ is of the form $y^2(cx^3 + fy^3)$, $L = 0$, $M = 0$; when ϕ is of the form $y(bx^4 + fy^4)$, $N = 0$, $L = 0$; and when ϕ is of the form $ax^5 + ey^5$, $M = 0$, $N = 0$; and of course when ϕ is of the form $y^3(dx^2 + fy^2)$, $L = 0$, $M = 0$, $N = 0$; it being obviously true in general, as remarked by Mr Cayley, that when not less than half the roots of a function of two variables are equal, all its invariants must vanish together.

Consequently we shall still have all the first minors of

$$\begin{vmatrix} A, & B, & C \\ \alpha, & \beta, & \gamma \\ \beta, & \gamma, & \delta \end{vmatrix}$$

zero, although there is not even so much as a pair of equal roots in ϕ; $AC - B^2$ however, it will be observed, is not zero in this supposition.

The theory of Hessians, simple or bordered, may be regarded as one among the infinite diversity of applications of the principle of the plexus. Let U, V, W, &c. be any number of concomitants having the common system of variables $x, y \ldots z$. Let χ represent

$$x'\frac{d}{dx} + y'\frac{d}{dy} + \ldots + z'\frac{d}{dz},$$

and take

$$\chi^2 U + \lambda\chi V + \text{\&c.} + \mu\chi W = S;$$

then

$$\frac{dS}{dx'}, \frac{dS}{dy'} \cdots \frac{dS}{dz'}$$

forms a plexus; and this, combined with χV, &c. ... χW, enables us to eliminate dialytically x', y', z', $\lambda \ldots \mu$. The result is a Hessian of U, bordered with

$$\frac{dV}{dx}, \frac{dV}{dy} \cdots \frac{dV}{dz}$$

horizontally and vertically, and also with

$$\frac{dW}{dx}, \frac{dW}{dy} \cdots \frac{dW}{dz},$$

&c. &c.

similarly dispersed; which Hessian, so bordered, is thus seen to be a concomitant to U, $V \ldots W$. The Hessian, as ordinarily bordered with ξ, $\eta \ldots \zeta$, is derived by taking for V the universal concomitant

$$x\xi + y\eta + \ldots + z\zeta,$$

and for W (if there be a double border)

$$x\xi' + y\eta' + \ldots + z\zeta',$$

and so forth.

If V be taken identical with U, the resulting form, consisting of U bordered with $\frac{dU}{dx}, \frac{dU}{dy} \cdots \frac{dU}{dz}$, has been shown* in my paper "On certain general Properties of Homogeneous Functions," in this *Journal*, to be equal to the product of the simple Hessian of U and of U itself multiplied by a

[* p. 173 above.]

numerical factor. The theory of the bordered Hessian may be profitably extended by taking

$$S = \chi^{2r} U + \lambda \chi^r V + \ldots + \mu \chi^r W,$$

and combining with $\chi^r V \ldots \chi^r W$ the plexus obtained by operating upon S with the rth powers and products of $\frac{d}{dx}, \frac{d}{dy} \cdots \frac{d}{dz}$, and eliminating dialytically the rth powers and products of $x', y' \ldots z'$. Thus if

$$U = ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4 \text{ and } V = (x\xi + y\eta)^2,$$

we obtain, by taking $S = \chi^4 U + \lambda \chi^2 V$, and proceeding as indicated in the preceding,

$$\begin{vmatrix} a, & b, & c, & \xi^2 \\ b, & c, & d, & \xi\eta \\ c, & d, & e, & \eta^2 \\ \xi^2, & \xi\eta, & \eta^2, & \end{vmatrix}$$

as a concomitant to U. So again, if

$$U = ax^5 + 5bx^4y + \ldots + fy^5,$$

we find

$$\begin{vmatrix} ax + by, & bx + cy, & cx + dy, & \xi^2 \\ bx + cy, & cx + dy, & dx + ey, & \xi\eta \\ cx + dy, & dx + ey, & ex + fy, & \eta^2 \\ \xi^2, & \xi\eta, & \eta^2, & \end{vmatrix}$$

a concomitant to U.

These extensions of the ordinary theory of Hessians will be found to be of considerable practical importance in the treatment of forms, for which reason they are here introduced.

Section VI. *On the Partial Differential Equations to Concomitants, Orthogonal and Plagiogonal Invariants, &c.*

In the 7th note of the Appendix to the three preceding sections* I alluded to the partial differential equations by which every invariant may be defined.

This method may also be extended to concomitants generally. M. Aronhold, as I collect from private information, was the first to think of the application of this method to the subject; but it was Mr Cayley who communicated to me the equations which define the invariants of functions of

[* p. 326 above.]

two variables*. The method by which I obtain these equations and prove their sufficiency is my own, but I believe has been adopted by Mr Cayley in a memoir about to appear in *Crelle's Journal.* I have also recently been informed of a paper about to appear in *Liouville's Journal* from the pen of M. Eisenstein, where it appears the same idea and mode of treatment have been made use of. Mr Cayley's communication to me was made in the early part of December last, and my method (the result of a remark made long before) of obtaining these and the more general equations, and of demonstrating their sufficiency, imparted a few weeks subsequently—I believe between January and February of the present year.

The method which I employ, in fact, springs from the very conception of what an invariant means, and does but throw this conception into a concise analytical form.

Suppose, to fix the ideas,

$$\phi = ax^n + nbx^{n-1}y + \tfrac{1}{2}n(n-1)cx^{n-2}y^2 + \ldots + ly^n,$$

and let $I(a, b, c \ldots l)$ be any invariant to ϕ.

Now suppose x to become $x + ey$, but y to remain unchanged; the modulus of the transformation, $\begin{vmatrix} 1, e \\ 0, 1 \end{vmatrix}$, being unity, I cannot alter in consequence of this substitution; but the effect of this substitution is to convert ϕ into the form

$$\alpha x^n + n\beta x^{n-1}y + \tfrac{1}{2}n(n-1)\gamma x^{n-2}y^2 + \ldots + \lambda y^n,$$

where $$\alpha = a, \quad \beta = b + ae, \quad \gamma = c + 2be + ae^2, \text{ \&c. \&c.}$$

$$\lambda = l + \ldots + nbe^{n-1} + ae^n.$$

Consequently, if we make

$$\Delta b = ae, \quad \Delta c = 2be + ae^2, \text{ \&c. \&c.},$$

we have by Taylor's theorem, observing that $\Delta a = 0$,

$$\Delta I = \left(\Delta b \frac{d}{db} + \Delta c \frac{d}{dc} + \text{\&c.}\right) I + \frac{1}{1.2}\left(\Delta b \frac{d}{db} + \Delta c \frac{d}{dc} + \text{\&c.}\right)^2 I$$
$$+ \frac{1}{1.2.3}\left(\Delta b \frac{d}{db} + \text{\&c.}\right)^3 I + \text{\&c.} = 0;$$

* It is extremely desirable to know whether M. Aronhold's equations are the same in form as those here subjoined. It is difficult to imagine what else they can be in substance. Should these pages meet the eye of that distinguished mathematician he will confer a great obligation on the author and be rendering a service to the theory by communicating with him on the subject: and I take this opportunity of adding that I shall feel grateful for the communication of any ideas or suggestions relating to this new Calculus from any quarter and in any of the ordinary mediums of language—French, Italian, Latin or German, provided that it be in the Latin character.

and this being true for all the values of e, every separate coefficient of e in ΔI must be zero: hence we obtain n different equations by equating to zero the coefficients of $e, e^2 \dots e^n$ respectively. The first of these equations will be

$$\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)\phi=0,$$

and it is obvious that this will imply all the rest; for, when e is taken indefinitely small, $I(a, b, c \dots)$ does not alter (when this equation is satisfied) by changing $a, b, c \dots$ into $a', b', c' \dots$; consequently $I(a', b', c', \&\text{c.})$ will not alter, when in place of a', b', c' we write a'', b'', c'', &c., obtained from a', b', c', &c., by the same law as a', b', c', &c., from a, b, c, &c.

Thus we may go on giving an indefinite number of increments, ey to x, without changing the value of I. Consequently, if the equation above written be satisfied, *à priori* all the rest must be so too. But there is not any difficulty in showing the same thing by a direct method*.

For we have

$$\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)I=0,$$

an identical equation. Hence

$$\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)\left\{\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)I\right\}=0;$$

hence

$$\left\{\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)\right\}I$$
$$+\left\{a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right\}^2 I=0,$$

that is

$$\left\{2\left(a\frac{d}{dc}+3b\frac{d}{dd}+6c\frac{d}{de}+\&\text{c.}\right)+\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)^2\right\}I=0;$$

repeating the application of the symbolic operator

$$\left(a\frac{d}{db}+2b\frac{d}{dc}+\&\text{c.}\right),$$

* The method above given has the advantage however of being immediately applicable to every species of concomitant, and we learn from it that concomitance, whether absolute or conditional, is sufficiently determined when affirmed to exist for *infinitesimal* variations; it cannot exist for infinitesimal variations without, by *necessary implication*, existing for finite variations also; a most important consideration this in conducing to a true idea of the nature of invariance and the other kinds of concomitance, and in cutting off all superfluous matter from the statement of the conditions by which they are defined.

we obtain

$$\left.\begin{aligned}&1.2.3\left\{a\frac{d}{dd}+4b\frac{d}{de}+10c\frac{d}{df}+\&\text{c.}\right\}\\&+1.2\left\{a\frac{d}{db}+2b\frac{d}{dc}+\&\text{c.}\right\}\left\{a\frac{d}{dc}+3b\frac{d}{dd}+\&\text{c.}\right\}\\&+\left(a\frac{d}{db}+2b\frac{d}{dc}+3c\frac{d}{dd}+\&\text{c.}\right)^3\end{aligned}\right\} I=0,$$

and so on; the numerical multipliers of the terms of the several series within the parentheses forming the regular succession of figurate numbers

$$\begin{matrix}1, & 2, & 3, \&\text{c.}\\ 1, & 3, & 6, \&\text{c.}\\ 1, & 4, & 10, \&\text{c.}\end{matrix}$$

It is easy to see that these equations correspond to the results of making the coefficients of the successive powers of e equal to zero.

I may remark, that the first instance as far as I know on record of this, (as some may regard it rather bold) but in point of fact perfectly safe and legitimate method of differentiating conjointly operator and operand, occurs in a paper by myself in this *Journal*, Feb. 1851, "On certain General Properties of Homogeneous Functions" [p. 165 above]; where I have applied it in operating with

$$\left\{(x_1-a_1e)\frac{d}{da_1}+(x_2-a_2e)\frac{d}{da_2}+\&\text{c.}\right\}$$

upon

$$\left\{(x_1-a_1e)\frac{d}{da_1}+(x_2-a_2e)\frac{d}{da_2}+\&\text{c.}\right\}^r\omega,$$

which, as I have there noticed, gives the result

$$\left\{(x_1-a_1e)\frac{d}{da_1}+\&\text{c.}\right\}^{r+1}\omega$$

$$-re\left\{(x_1-a_1e)\frac{d}{da_1}+\&\text{c.}\right\}^r\omega.$$

The equation $\left(a\frac{d}{db}+2b\frac{d}{dc}+\&\text{c.}\right)I=0$ is evidently not enough to define I as an invariant; it merely serves to show that I does not alter when in place of x we write $x+ey$, but this is true for any function of the differences of the roots of the form multiplied by a suitable power of a, namely that power which is just sufficient to cause the product to become integer. But if we now, for convenience, write

$$\phi=ax^n+nbx^{n-1}y+\tfrac{1}{2}n(n-1)cx^{n-2}y^2+\dots$$
$$+\tfrac{1}{2}n(n-1)c'x^2y^{n-2}+nb'xy^{n-1}+a'y^n,$$

and form the similar equation from the other side, namely

$$\left(a'\frac{d}{db'} + 2b'\frac{d}{dc'} + 3c'\frac{d}{dd'} + \&c.\right) I = 0,$$

these two equations together will suffice to define any invariant, as I shall proceed to show—these are the two equations alluded to brought under my notice by Mr Cayley. If they coexist, it follows from the method by which I have deduced them that x may be changed into $x + ey$, or y into $y + fx$, without I being altered, e and f having any values whatever: and it is obvious that these substitutions may be performed, not merely alternatively but successively, because the equations between the coefficients are identical equations, and depend only on the form of I.

Let now x become $x + ey$, and then y become $y + fx$; the result of these substitutions is to convert

$$x \text{ into } x + efx + ey,$$

and

$$y \text{ into } \quad fx + y.$$

Finally, let x become $x + gy$; then x is converted into $(1 + ef)(x + gy) + ey$, and y into $y + f(x + gy)$,

that is

$$x \text{ becomes } (1 + ef)x + (eg + efg)y,$$

and

$$y \text{ becomes } \quad fx + \ (1 + fg)y.$$

The modulus of substitution it is evident, *à priori*, always remains unity, and nothing would be gained by pushing the substitutions any further, as it is clear that we may satisfy the equations

$$1 + ef = p, \quad e + g + efg = q,$$
$$f = p', \qquad 1 + fg = q',$$

for all values of p, q, p', q', which satisfy the equation

$$pq' - p'q = 1,$$

and for none other except such values; hence I remains unaltered for any unit-modular linear transformation of x, y, and is therefore an invariant by definition.

If ϕ be taken a function of three variables, x, y, z, and be thrown under the form

$$az^n + (a_1x + b_1y)z^{n-1} + (a_2x^2 + 2b_2xy + c_2y^2)z^{n-2} + \&c.,$$

and I be any invariant of ϕ, by supposing x to become $x + ey$, and giving b_1, b_2, c_2, &c., the corresponding variations, and taking e indefinitely small, we obtain

$$\left\{a_1\frac{d}{db_1} + \left(a_2\frac{d}{db_2} + 2b_2\frac{d}{dc_2}\right) + \left(a_3\frac{d}{db_3} + 2b_3\frac{d}{dc_3} + 3c_3\frac{d}{dd_3}\right) + \&c.\right\} I = 0,$$

$$\left\{b_1\frac{d}{da_1} + \left(c_2\frac{d}{db_2} + 2b_2\frac{d}{da_2}\right) + \&c.\ \&c.\right\} I = 0:$$

and in like manner, by arranging ϕ according to the powers of y and of x, we obtain two other pairs of equations: it is clear, however, that three equations (it would seem any three out of the six) would suffice and imply the other three. The method of demonstration will be the same as in the instance of two variables: First, it can be shown by the method of successive accretions, that I remaining invariable when x receives an indefinitely small increment ϵy, or y an indefinitely small increment ϵz, or z an indefinitely small increment ϵx, it will also remain invariable when these increments are taken of any finite magnitude. Secondly, by eight successive transformations, admissible by virtue of the preceding conclusion, x, y, z may be changed into any linear functions of x, y, z, consistent with the modulus of transformation being unity. And in general for a function of m variables, m partial differential equations similarly constructed (but not however arbitrarily selected) will be necessary and sufficient to determine any invariant: and it is clear that all the general properties of invariants must be contained in and be capable of being educed out of such equations.

The same method enables us also to establish the partial differential equations for any covariant, or indeed any concomitant whatever.

Thus let

$$\phi = ax^n + nbx^{n-1}y + \tfrac{1}{2}n(n-1)cx^{n-2}y^2 + \ldots + nb'xy^{n-1} + a'y^n = 0,$$

and let $K(a, b, c, \&c.;\ x, y, x', y', \&c.;\ \xi, \eta, \&c.)$ represent any concomitant, $x, y;\ x', y'$ being cogredient, and ξ, η, &c. contragredient systems; when x, y become $x+ey, y$, any such system x', y' becomes $x'+ey', y'$; and any such system as ξ, η becomes $\xi, \eta - e\xi$; and taking e indefinitely small, the second coefficients a, b, c, &c. become $a, b+ae, c+2be$, &c. as before; hence the equation to the concomitant becomes

$$\left\{a\frac{d}{db} + 2b\frac{d}{dc} + \ldots - y\frac{d}{dx} - y'\frac{d}{dx'} + \ldots + \xi\frac{d}{d\eta} - \&c.\right\} = 0^*;$$

and in like manner, by changing y into $y+ex$, results the corresponding equation

$$\left\{a'\frac{d}{db'} + 2b'\frac{d}{dc'} + \ldots - x\frac{d}{dy} - x'\frac{d}{dy'} + \ldots + \eta\frac{d}{d\xi} - \&c.\right\} K = 0.$$

These two equations define in a perfectly general manner every concomitant (with any given number of cogredient and contragredient systems) to the form ϕ; and the due number of pairs of similarly constituted equations will serve to define the concomitant to a function of any given number of variables†.

* For we have

$$K(a, b+ae, c+2be, \&c.;\ x, y, \&c.;\ \xi, \eta, \&c.)$$
$$= K(a, b, c, \&c.;\ x, x+ey, \&c.;\ \xi, \eta - e\xi, \&c.;\ \&c.).$$

† *Vide* Note (10) [p. 361 below].

In like manner we may proceed to form the equations corresponding to what may be termed *conditional* concomitants, whether *orthogonal* or *plagiogonal*. The concomitants previously considered may be termed absolute, the linear transformations admissible being independent of any but the one general relation, imposed merely for the purpose of convenience, namely of their modulus being made unity. An orthogonal concomitant is a form which remains invariable, not for arbitrary unit-modular, but for orthogonal transformation, that is for linear substitutions of $x, y \ldots z$, which leave unchanged $x^2 + y^2 + \ldots + z^2$: in like manner, a plagiogonal concomitant may be defined of a form which remains invariable for all linear substitutions of $x, y \ldots z$, which leave unaltered any given quadratic function of $x, y \ldots z$. Thus, let it be required to express the condition of $Q(a, b, c \ldots x, y; \xi, \eta)$, being an orthogonal concomitant to the form

$$ax^n + nbx^{n-1}y + \ldots + nb'xy^{n-1} + a'y^n.$$

Let x become $x + ey$, e being indefinitely small, then y must become $y - ex$, and the variations of $a, b \ldots b', a'$ will be the sum of the variations produced by taking separately $x + ey$ for x and $y - ex$ for y. Hence the one sole condition for Q being of the required form becomes

$$\left.\begin{cases} \left(a\dfrac{d}{db} + 2b\dfrac{d}{dc} + \ldots - y\dfrac{d}{dx} + \xi\dfrac{d}{d\eta}\right) \\ -\left(a'\dfrac{d}{db'} + 2b'\dfrac{d}{dc'} + \ldots - x\dfrac{d}{dy} + \eta\dfrac{d}{d\xi}\right) \end{cases}\right\} Q = 0,$$

or, as it may be written, $\theta Q - \omega Q = 0$, where $\theta Q = 0$, $\omega Q = 0$ are the two equations expressing the conditions of Q, being an unconditional or absolute concomitant; and so in general if ϕ be a function of m variables, we may obtain $\frac{1}{2}m(m-1)$ equations of the form $L - M = 0$ for the concomitant, of which however $(m-1)$ only will be independent.

Supposing, again, the substitutions to which x, y are subject to be conditioned by $lx^2 + 2mxy + ny^2$ remaining unalterable, or which is a more convenient and only in appearance less general supposition by $x^2 + 2mxy + y^2$ remaining unalterable, the general type of an infinitesimal system of substitutions will be rendered by supposing x, y to become $(1 + me)x + ey$, $-ex + (1 - me)y$, respectively, for then $x^2 + 2mxy + y^2$ becomes

$$(1 - m^2e^2)x^2 + \{2m + (2m - 2m^2)e^2\}xy + (1 - m^2e^2)y^2,$$

which differs from $x^2 + 2mxy + y^2$ only by quantities of the second order of smallness which may be neglected, and ξ and η will therefore become $(1 - me)\xi - e\eta$, $-ex + (1 + me)y$, respectively: then, as to the coefficients of ϕ, in addition to the variations which they undergo when m is zero, there will be the variations consequent upon x assuming the increment mex, and y

the increment $-mey$: but by making x become $x+mex$, a, b, c, &c., b', a' assume respectively the variations

$$n\,.\,mea,\ (n-1)\,meb,\ \ldots\ meb',\ 0,\ \text{respectively};$$

and by making y become $y-mey$, the corresponding variations become

$$0,\ -meb,\ \ldots -(n-1)\,meb',\ -n\,.\,mea',\ \text{respectively}.$$

Hence the equation becomes

$$\theta Q-\omega Q+m\,(\lambda Q-\mu Q)=0,$$

where θ and ω have the same signification as before, and where λ denotes

$$na\frac{d}{da}+(n-1)\,b\frac{d}{db}+\ldots+b'\frac{d}{db'}+x\frac{d}{dx}-\xi\frac{d}{d\xi},$$

and μ denotes

$$b\frac{d}{db}+2c\frac{d}{dc}+\ldots\ldots\ldots+na'\frac{d}{da'}-y\frac{d}{dy}+\eta\frac{d}{d\eta}.$$

If there be several systems of x, y or of ξ, η, or of both, the only difference in the equation of condition will consist in putting

$$\Sigma\left(y\frac{d}{dx}\right),\quad \Sigma\left(x\frac{d}{dy}\right),\quad \Sigma\left(x\frac{d}{dx}\right),\quad \Sigma\left(y\frac{d}{dy}\right),$$

$$\Sigma\left(\eta\frac{d}{d\xi}\right),\quad \Sigma\left(\xi\frac{d}{d\eta}\right),\quad \Sigma\left(\xi\frac{d}{d\xi}\right),\quad \Sigma\left(\eta\frac{d}{d\eta}\right),$$

instead of the single quantities included within the sign of definite summation.

Fearing to encroach too much on the limited space of the *Journal*, I must conclude for the present with showing how to integrate the general equation to the orthogonal invariant of ϕ, the general function of x, y.

Beginning with $\phi=ax^2+2bxy+cy^2$, the equation becomes

$$\left\{-2b\frac{d}{da}+(a-c)\frac{d}{db}+2b\frac{d}{dc}+y\frac{d}{dx}-x\frac{d}{dy}\right\}Q=0.$$

Write now

$$da=-2bd\theta,\qquad dx=yd\theta,$$
$$db=(a-c)\,d\theta,\quad dy=-xd\theta,$$
$$dc=+2bd\theta;$$

we have then

$$\lambda da+\mu db+\nu dc=d\theta\,\{\mu a+2(\nu-\lambda)\,b-\mu c\}.$$

Let $\qquad \mu=\kappa\lambda,\quad 2(\nu-\lambda)=\kappa\mu,\quad -\mu=\kappa\nu;$

then $\qquad d\log(\lambda a+\mu b+\nu c)=\kappa d\theta;$

or $\qquad \lambda a+\mu b+\nu c=be^{\kappa\theta}.$

To find κ we have the determinant

$$\begin{vmatrix} \kappa, & -1, & 0 \\ 2, & \kappa, & -2 \\ 0, & 1, & \kappa \end{vmatrix} = 0,$$

that is, $$\kappa^3 + 4\kappa = 0,$$

and calling the three roots of this equation κ_1, κ_2, κ_3, we have

$$\kappa_1 = 0, \quad \kappa_2 = 2\iota, \quad \kappa_3 = -2\iota;$$

accordingly we may put

$$\kappa = 0, \quad \lambda = 1, \quad \mu = 0, \quad \nu = 1,$$

or $$\kappa = 2\iota, \quad \lambda = 1, \quad \mu = 2\iota, \quad \nu = -1,$$

or $$\kappa = -2\iota, \quad \lambda = 1, \quad \mu = -2\iota, \quad \nu = -1.$$

Again, $$pdx + qdy = (py - qx)\,d\theta;$$

and putting $-q = ep$, $p = eq$, so that $px + qy = Ee^{e\theta}$,

$$e^2 = -1, \quad e_1 = \iota, \quad e_2 = -\iota;$$

and we may put

$$e = \iota, \quad p = 1, \quad q = -\iota,$$

or $$e = -\iota, \quad p = 1, \quad q = +\iota.$$

Consequently the complete integral of the given partial differential equation is found by writing

$$a + c = l, \qquad x - \iota y = Ee^{\iota\theta},$$

$$a + 2\iota b - c = l'e^{2\iota\theta}, \quad x + \iota y = E'e^{-\iota\theta},$$

$$a - 2\iota b - c = l''e^{-2\iota\theta}.$$

By means of these five equations, after eliminating θ, we may obtain four independent equations between a, b, c; x, y. Suppose

$$Q_1 = 0, \quad Q_2 = 0, \quad Q_3 = 0, \quad Q_4 = 0;$$

then $Q = F(Q_1, Q_2, Q_3, Q_4)$ is the complete integral required.

Pursuing precisely the same method for the general case, it will be found that, calling the degree of the given function n when n is even, the equation in κ to be solved will be

$$\kappa(\kappa^2 + 4)(\kappa^2 + 9)\ldots(\kappa^2 + n^2) = 0;$$

and when n is odd (say $2m + 1$), the equation in κ to solve will be

$$(\kappa + 1)(\kappa^2 + 9)\ldots(\kappa^2 + n^2) = 0;$$

and performing the necessary reductions, and calling the roots of the equation, arranged in order of magnitude, $\kappa_1\iota$, $\kappa_2\iota$ ... $\kappa_n\iota$, respectively, it will be found that the equations containing the integral become

$$\left.\begin{array}{l} L_1 = l_1 e^{\kappa_1\iota\theta} \\ L_2 = l_2 e^{\kappa_2\iota\theta} \\ L_3 = l_3 e^{\kappa_3\iota\theta} \\ \dots\dots\dots\dots\dots\dots \\ L_{n+1} = l_{n+1} e^{\kappa_{n+1}\iota\theta} \end{array}\right\} \quad \left.\begin{array}{l} x - \iota y = Ee^{\iota\theta} \\ x + \iota y = E'e^{-\iota\theta} \end{array}\right\},$$

where $l_1, l_2 \dots l_{n+1}$; E, E' are arbitrary constants, and where $L_1, L_2 \dots L_{n+1}$ are the values assumed by the 1st, 2nd ... $(n+1)$th coefficients of the given function ϕ, or

$$ax^n + nbx^{n-1}y + \dots + nb'xy^{n-1} + a'y^n,$$

when it is transformed by writing $x+\iota y$ in place of x, and $y+\iota x$ in place of y. ι is of course employed in the foregoing according to the usual notation to represent $\surd(-1)$. The same method applies to the general theory of plagiogonal concomitants, where the linear substitutions are supposed such as to leave $lx^2+2mxy+ny^2$ unaltered in form, and the equations in θ which contain the integral present themselves under a similar aspect. But a more full discussion of these interesting integrals must be reserved until the ensuing number of the *Journal*.

NOTES IN APPENDIX.

(9) The scale of covariants to a function of (x, y) obtained by the method of unravelment [on p. 297 above], may be otherwise deduced in a form more closely analogous to that of the corresponding theorems for the corresponding invariantive scale [on p. 295 above], by a method which has the advantage of exhibiting the scale equally well for the case of functions of the degree $4\iota+2$ or $4\iota+4$, the only difference being that in the latter case the coefficients of the odd powers of λ will be found all to vanish, so that the degrees of the covariants will rise by steps of 4 instead of by steps of 2, just conversely to what happens in the invariantive scale; whereas in the invariantive scale alluded to the forms containing odd powers of λ vanish when the degree of the function is of the form $4\iota+2$, but do not vanish when it is of the form 4ι. This method in the form here subjoined is a slight modification of one suggested to me by my friend Mr Cayley.

Let F be the given function of x, y of the degree $2n$; take the systems x', y'; x_1, y_1 cogredient with one another and with x, y. Then form the concomitant

$$K = \left(x'\frac{d}{dx} + y'\frac{d}{dy}\right)^n F + \lambda\,(x'y - y'x)^{n-1}(x'y_1 - y'x_1)(xy_1 - yx_1).$$

Then (by what may be termed the Divellent method, which has been previously applied by me in the *Philosophical Magazine* for Nov. 1851) calling $\theta_0, \theta_1, \theta_2 \ldots \theta_n$, the coefficients of

$$x'^n,\ x'^{n-1},\ y',\ \ldots\ y'^n \text{ in } K,$$

we shall have

$$\theta_0 = A_0 x^n + B_0 x^{n-1} y + \ldots + L_0 y^n,$$
$$\theta_1 = A_1 x^n + B_1 x^{n-1} y + \ldots + L_1 y^n,$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\theta_n = A_n x^n + B_n x^{n-1} y + \ldots + L_n y^n,$$

the coefficients being functions of the coefficients of f and of quadratic combinations of x_1, y_1, affected with the multiplier λ; and the determinant

$$\begin{vmatrix} A_0, & B_0 \ldots L_0 \\ A_1, & B_1 \ldots L_1 \\ \ldots & \ldots\ldots \\ A_n, & B_n \ldots L_n \end{vmatrix}$$

will give a function of λ in which the coefficients of the several powers of λ will be all zero or covariants of F.

The actual form of this determinant is not here given for want of space and time, but will be exhibited hereafter. Precisely an analogous method applies to obtain the scale to $(x, y, z)^4$ given in Note (2) [p. 322 above]. Calling $F = (x, y, z)^4$, let the systems x', y', z'; x_1, y_1, z_1, be taken cogredient with one another and with x, y, z. Then, using R to express the determinant

$$\begin{vmatrix} x', & y', & z' \\ x, & y, & z \\ x_1, & y_1, & z_1 \end{vmatrix},$$

and making

$$K = \left(x' \frac{d}{dx} + y' \frac{d}{dy} + z' \frac{d}{dz}\right)^2 F + \lambda R,$$

and proceeding as above by the divellent method, we obtain the scale required.

(10) [p. 356 above.] It is obvious that these defining equations ought to give the means of discovering and verifying all the properties of concomitants; but it is very difficult to see how in the present state of analysis many of the general theorems that have been stated, readily admit of being deduced from them.

The comparatively simple but eminently important theory of the evector symbol does however admit of a very pretty verification by aid of these equations. Thus, suppose θ any concomitant; suppose a contravariant to a function F of x, y, say

$$ax^n + nbx^{n-1}y + \ldots + nb'xy^{n-1} + a'y^n.$$

Then θ must satisfy the two equations

$$\left(L+\xi\frac{d}{d\eta}\right)\theta=0,\quad \left(L'+\eta\frac{d}{d\xi}\right)\theta=0,$$

where
$$L=a\frac{d}{db}+2b\frac{d}{dc}+\ldots+nb'\frac{d}{da'},$$

$$L'=a'\frac{d}{db'}+2b'\frac{d}{dc'}+\ldots+nb\frac{d}{da}.$$

Now let $\phi=\chi(\theta)$ where

$$\chi=\xi^n\frac{d}{da}+\xi^{n-1}\eta\frac{d}{db}+\xi^{n-2}\eta^2\frac{d}{dc}+\ldots+\eta^n\frac{d}{da'};$$

then
$$L(\chi\theta)=\chi(L\theta)-(\chi L)\theta$$

$$=\chi(L\theta)-\left(\xi^n\frac{d}{db}+2\xi^{n-1}\eta\frac{d}{dc}+\ldots+n\xi\eta^{n-1}\frac{d}{da'}\right)\theta,$$

$$\xi\frac{d}{d\eta}(\chi\theta)=\chi\left(\xi\frac{d}{d\eta}\theta\right)+\left(\xi\frac{d}{d\eta}\chi\right)\theta$$

$$=\chi\left(\xi\frac{d}{d\eta}\theta\right)+\left(\xi^n\frac{d}{db}+2\xi^{n-1}\eta\frac{d}{dc}+\ldots+n\xi\eta^{n-1}\frac{d}{da'}\right)\theta.$$

Hence
$$\left(L+\xi\frac{d}{d\eta}\right)\chi(\theta)=\chi\left\{\left(L+\xi\frac{d}{d\eta}\right)\theta\right\}=\chi(0)=0.$$

Similarly
$$\left(L'+\eta\frac{d}{d\xi}\right)\chi(\theta)=0.$$

Hence if θ is an integral of the two conditioning equations, so also is $\chi(\theta)$. In like manner, if θ be a covariant or any other kind of concomitant of F, it may be proved that its evectant $\chi(\theta)$ is the same.

(11) [p. 331 above.] Very much akin with the supposed equations is the following most remarkable equation, which can be proved to exist. Let ϕ be a function of x and y of the 5th degree. Let P and Q be the quadratic and cubic covariants of ϕ. P is of two dimensions in the coefficients and also in the variables, and Q of three dimensions in both; they are in fact the s and t (in respect to x' and y') of $\left(x'\frac{d}{dx}+y'\frac{d}{dy}\right)^4\phi$. Then, giving P and Q proper numerical factors, it will be found that

$$H_2\phi+PH\phi+Q\phi=0.$$

I believe that a similar equation connects any function of x and y above the 3rd degree with its first and second Hessians. The proof will be given in a subsequent Section, where also I shall give a complete proof, which occurred to me immediately after sending the preceding note to the press, of the complete Theory of the Respondent by means of the general equations of concomitance.

P.S. Since the preceding was in type, I have ascertained the existence and sufficiency of a general method for forming the polar reciprocal and probably also the discriminant to functions of any degree of three variables by an explicit process of permutation and differentiation. In particular I am enabled to give the actual rule for constructing the polar reciprocal and the discriminant curves of the 4th and 5th degrees. So far as regards the polar reciprocal of curves of the 4th degree M. Hesse has already given a method of obtaining it, but mine is entirely unlike to this, and rests upon certain extremely simple and universal principles of the calculus of forms. The only thing necessary to be done in order to carry on the process to curves of the 6th or higher degrees, is to ascertain the relation of the discriminants of functions of two variables of those respective degrees to such of the fundamental invariants as are of an inferior order to the discriminant.

The theory applies equally well to surfaces and to functions of any number of variables, and may, I believe, without any serious difficulty be extended so as to reduce to an explicit process the general problem of effecting the elimination between functions of any degree and of any number of variables. The method above adverted to will appear in a subsequent Section.

[*Continued pp.* 402 *and* 411 *below.*]

44.

SUR UNE PROPRIÉTÉ NOUVELLE DE L'ÉQUATION QUI SERT A DÉTERMINER LES INÉGALITÉS SÉCULAIRES DES PLANÈTES.

[*Nouvelles Annales de Mathématiques*, XI. (1852), pp. 438—440.]

[Extract.]

6. *Soit le déterminant carré symétrique*

$$\begin{vmatrix} a_{1,1}, & a_{1,2} \dots a_{1,n} \\ a_{2,1}, & a_{2,2} \dots a_{2,n} \\ \dots & \dots \\ a_{n,1}, & a_{n,2} \dots a_{n,n} \end{vmatrix}, \qquad \text{(M)}$$

dans lequel on a, d'après la définition,

$$a_{l,c} = a_{c,l}.$$

Élevant le déterminant à la puissance p, on obtient le déterminant

$$\begin{vmatrix} A_{1,1}, & A_{1,2} \dots A_{1,n} \\ A_{2,1}, & A_{2,2} \dots A_{2,n} \\ \dots & \dots \\ A_{n,1}, & A_{n,2} \dots A_{n,n} \end{vmatrix}; \qquad \text{(N)}$$

et ce déterminant est symétrique aussi par rapport à la diagonale $A_{1,1}$, $A_{2,2} \dots A_{n,n}$.

Retranchant de chaque terme de la diagonale symétrique de (M) *la même quantité* λ, *on obtient le déterminant*

$$\begin{vmatrix} a_{1,1}-\lambda, & a_{1,2} \dots a_{1,n} \\ a_{2,1}, & a_{2,2}-\lambda \dots a_{2,n} \\ \dots & \dots \\ a_{n,1}, & a_{n,2} \dots a_{n,n}-\lambda \end{vmatrix}. \qquad \text{(P)}$$

Développant ce déterminant et ordonnant par rapport à λ, *on obtient une expression qui, étant égalée à zéro, donne l'équation*

$$\lambda^n - f\lambda^{n-1} + g\lambda^{n-2} + \dots (-1)^n t = 0, \qquad (1)$$

équation qui a n racines réelles (voir t. x. p. 259).

Retranchant de chaque terme de la diagonale symétrique du déterminant (N) *la quantité* μ, *et opérant comme ci-dessus, on parvient à l'équation*

$$\mu^n - F\mu^{n-1} + G\mu^{n-2} + \dots (-1)^n T = 0, \qquad (2)$$

équation qui a aussi n racines réelles. Les racines de cette équation sont les racines de l'équation (1), *élevées chacune à la puissance p.*

Démonstration. Représentons par

$$\rho_1, \quad \rho_2, \quad \rho_3 \cdots \rho_p,$$

les p racines de l'équation $\rho^p - 1 = 0$. Écrivons le déterminant

$$\begin{vmatrix} a_{1,1} - \rho_q\lambda, & a_{1,2} \cdots a_{1,n} \\ a_{2,1}, & a_{2,2} - \rho_q\lambda \cdots a_{2,n} \\ \cdots\cdots & \cdots\cdots \\ a_{n,1} \cdots\cdots & a_{n,n} - \rho_q\lambda \end{vmatrix},$$

et faisons q égal successivement à tous les nombres de la suite $1, 2, 3 \dots p$, on aura p déterminants; le produit de tous ces déterminants reste évidemment le même dans quelque ordre qu'on prenne ces déterminants, et, d'après les propriétés connues des racines de l'unité, tous les termes en ρ qui ne seront pas élevés à une puissance p disparaîtront, et λ accompagnant toujours ρ, il ne reste donc que des λ^p, et le déterminant-produit sera

$$\begin{vmatrix} A_{1,1} - \lambda^p, & A_{1,2}, & A_{1,3} \dots A_{1,n} \\ A_{2,1}, & \dots, & A_{2,2} - \lambda^p \dots A_{2,n} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots \\ A_{n,1}, & A_{n,2}, & \dots\dots A_{n,n} - \lambda^p \end{vmatrix}; \qquad (Q)$$

où, faisant abstraction de λ, on a le déterminant (N). Ainsi

$$\mu = \lambda^p.$$

C. Q. F. D.

7. *Application.*

$$n = 2, \text{ et } p = 2;$$

déterminant
$$\begin{vmatrix} a, & b \\ b, & c \end{vmatrix}, \qquad (M)$$

élevant ce déterminant au carré, on a

$$\begin{vmatrix} a^2 + b^2, & ab + bc \\ ab + bc, & b^2 + c^2 \end{vmatrix}; \qquad (N)$$

déterminant
$$\begin{vmatrix} a-\lambda, & b \\ b, & c-\lambda \end{vmatrix}, \tag{P}$$

$$\lambda^2-(a+c)\lambda+ac-b^2=0; \tag{1}$$

déterminant
$$\begin{vmatrix} a^2+b^2-\mu, & ab+bc \\ ab+bc, & b^2+c^2-\mu \end{vmatrix},$$

$$\mu^2-(a^2+c^2+2b^2)\mu+(ac-b^2)^2=0, \text{ où } \mu=\lambda^2. \tag{2}$$

Faisons
$$n=2, \quad p=3,$$
(M) ne change pas, et l'on a
$$\begin{vmatrix} a^3+2ab^2+b^2c, & a^2b+abc+b^3+bc^2 \\ a^2b+abc+b^3+b^2c, & ab^2+2b^2c+c^3 \end{vmatrix}; \tag{N}$$
le déterminant (P) et l'équation (1) restent les mêmes; mais l'équation (2) devient
$$\mu^2-(a^3+c^3+3ab^2+3cb^2)\mu+(ac-b^2)^3=0,$$
où
$$\mu=\lambda^3,$$
car, λ_1 et λ_2 étant les deux racines de l'équation (1), on a
$$\lambda_1^3+\lambda_2^3=a^3+c^3+3ab^2+3cb^2, \quad \lambda_1^3\lambda_2^3=(ac-b^2)^3.$$

8. M. Sylvester fait observer que son théorème est un cas particulier d'un théorème plus général, démontré par M. Borchardt, pour des déterminants quelconques, et qui devient le théorème démontré ci-dessus, lorsque le déterminant est symétrique (*Journal de Mathématiques*, t. XII. p. 63, 1847).

45.

ON A REMARKABLE THEOREM IN THE THEORY OF EQUAL ROOTS AND MULTIPLE POINTS.

[*Philosophical Magazine,* III. (1852), pp. 375—378.]

IN order that the theorem which I propose to state may be the more easily understood, and with the least ambiguity expressed, I shall commence with the case of a homogeneous function of two variables only, x and y.

Let

$$\phi = ax^n + nbx^{n-1}y + \tfrac{1}{2}n(n-1)cx^{n-2}y^2 + \ldots + nb'xy^{n-1} + a'y^n,$$

and let the result of operating with the symbol

$$x^n \frac{d}{da} + x^{n-1}y\frac{d}{db} + \ldots + y^{n-1}x\frac{d}{db'} + y^n\frac{d}{da'},$$

on any function of $a, b, c \ldots b', a'$ be called the Evectant of such function, and the result of repeating this process r times the rth Evectant.

Understand by the multiplicity of the equation the number of equalities between the roots that exist; so that a pair of equal roots will signify a multiplicity 1, two pairs of equal roots, or three equal roots a multiplicity 2; a pair of equal roots and a set of three equal roots, a multiplicity $1 + 2$ or 3, and so on. Now suppose the total multiplicity of ϕ to be m: the first part of the proposition consists in the assertion that the 1st, 2nd, 3rd ... $(m-1)$th Evectants of the discriminant of ϕ, that is of the result of eliminating x and y between $\frac{d\phi}{dx}, \frac{d\phi}{dy}$ (as well as the discriminant itself), will all vanish in whatever way the multiplicity is distributed; the second part of the proposition about to be stated requires that the mode should be taken into account of the manner in which the multiplicity (m) is made up. Suppose, then, that there are r groups of roots, for one of which the multiplicity is m_1, for the second m_2, &c., and for the rth m_r, so that $m_1 + m_2 + \ldots + m_r = m$. Then, I say, that the mth evectant of the determinant of ϕ is of the form

$$(a_1x + b_1y)^{m_1n} (a_2x + b_2y)^{m_2n} \ldots (a_rx + b_ry)^{m_rn},$$

where $a_1 : b_1$, $a_2 : b_2 \ldots a_r : b_r$ are the ratios of $x : y$ corresponding to the several sets of equal roots.

This latter part of the theorem for the case of $m = 1$ was discovered inductively by Mr Cayley, by considering the cases when ϕ is a cubic, or a biquadratic function. I extended the theory to functions of any number of variables, and supplied a demonstration, that is for the case of one pair of equal roots. Mr Salmon showed that my demonstration could be applied to the case of two pairs of equal roots, or two double points, &c., and very nearly at the same time I made the like extension to the case of three equal roots, cusps, &c., and almost immediately after I obtained a demonstration for the theorem in its most general form. This demonstration reposes upon a very refined principle, which I had previously discovered but have not yet published, in the Theory of Elimination.

I have here anticipated a little in speaking of the theorem as applicable to curves and other loci.

Suppose $\phi(x, y, z) = 0$ to be the equation to a curve expressed homogeneously.

Let

$$\begin{aligned}\phi(x, y, z) = ax^n &+ (na'x^{n-1}y + nb'x^{n-1}z)\\ &+ \tfrac{1}{2}n(n-1)a''x^{n-2}y^2 + n(n-1)b''x^{n-2}yz + \tfrac{1}{2}n(n-1)c''x^{n-2}z^2,\\ &+ \&c. \qquad \&c.,\end{aligned}$$

and understand by the evectant of any quantity the result of operating upon it with the symbol

$$x^n \frac{d}{da} + x^{n-1}y\frac{d}{da'} + x^{n-1}z\frac{d}{db'} + x^{n-2}y^2\frac{d}{da''} + \&c.$$

Suppose, now, the curve to have double points, the $(r-1)$th evectant (and of course all the inferior evectants) of the discriminant of ϕ (meaning thereby the result of eliminating x, y, z between $\frac{d\phi}{dx}$, $\frac{d\phi}{dy}$, $\frac{d\phi}{dz}$) will all vanish, and the rth evectant will be of the form

$$(a_1x + b_1y + c_1z)^n \times (a_2x + b_2y + c_2z)^n \ldots \times (a_rx + b_ry + c_rz)^n,$$

where $a_1 : b_1 : c_1$, $a_2 : b_2 : c_2 \ldots a_r : b_r : c_r$ are the ratios of the coordinates at the respective double points. If there be cusps the multiplicity of each such will be 2; and calling the total multiplicity m, to every cusp will correspond a factor of the $2n$th power in the mth evectant; and so on in general for various degrees of multiplicity at the singular points respectively. The like theorem extends to conical and other singular points of surfaces; so that there exists a method, when a locus is given having any degree of multiplicity, of at once detecting the amount and distribution of this multiplicity, and the positions of the one or more singular points. In conclusion I may state, that precisely analogous results (*mutatis mutandis*) obtain, when, in place of a single function having multiplicity, we take the more general supposition of any number of homogeneous functions being subject to the condition of pluri-simultaneity, that is being capable of being made to vanish by each of several different systems of values for the ratios between the variables. Multiplicity in a single function is, in fact, nothing more nor less than pluri-simultaneity existing between the functions derived from it by differentiating with respect to each of the given variables successively. But as I purpose to give these theorems and their demonstration, which I have already imparted to my mathematical correspondents, in a paper destined for reading before the Royal Society, I need not further enlarge upon them on the present occasion.

P.S. In the above statement I have spoken only of cusps of curves which are the precise and unambiguous analogues of three coincident points in point-systems, in order to avoid the necessity of entering into any disquisition as to the species of singularity in curves or other loci corresponding to higher degrees of multiplicity in point-systems, a subject which has not hitherto been completely made out. I may here also add a remark, which gives a still higher interest to the theory, which is (to confine ourselves, for the sake of brevity, to functions of two variables), that if any root of $x:y$, say $a:b$, occur $1+\mu$ times, the total multiplicity of the equation being supposed m, and its degree n, then taking ι any integer number not exceeding μ, the $(m+\iota)$th evectant of the discriminant will contain the factor $(ax+by)^{(\mu-\iota)n}$. So that, for instance, if there be but a single group of equal roots, and they be $1+\mu$ in number, every evectant up to the $(\mu-1)$th inclusive will vanish, and from the μth to the $(2\mu-\iota)$th will contain a power of $(ax+by)^n$.

46.

OBSERVATIONS ON A NEW THEORY OF MULTIPLICITY.

[*Philosophical Magazine*, III. (1852), pp. 460—467.]

In the Postscript to my paper in the last number of the *Magazine*, I mis-stated, or to speak more correctly, I understated the law of Evection applicable to functions having any given amount of distributive multiplicity. The law may be stated more perfectly, and at the same time more concisely, as follows. Every point represented by the coordinates $\alpha_1, \beta_1 \ldots \gamma_1$, for which the multiplicity is m_1, will give rise in *every evectant** of the discriminant of the function to a factor $(\alpha_1 x + \beta_1 y + \ldots + \gamma_1 z)^{m_1 n}$, n being supposed to be the degree of the function. Hence if there be r such points, for which the several multiplicities are $m_1, m_2 \ldots m_r$, every evectant must contain $(m_1 + m_2 + \ldots + m_r) n$ linear factors; and as the ιth evectant is of the degree ιn, it follows that all the evectants below the $(m_1 + m_2 + \ldots + m_r)$th evectant must vanish completely, and this Evectant itself be contained as a factor in all above it†. When a function of only two variables is in question, there is no difficulty in understanding what property of the function it is which is indicated by the allegation of the existence of multiplicities $m_1, m_2 \ldots m_r$;

* Frequent use being made in what follows of the word Evectant, I repeat that the evectant of any expression connected with the coefficients of a given function (supposed to be expressed in the more usual manner with letters for the coefficients affected with the proper binomial or polynomial numerical multipliers) means the result of cperating upon such expressions with a symbol formed from the given function by suppressing all the binomial or polynomial numerical parts of the coefficients to be suppressed, and writing in place of the literal parts of the coefficients a, b, c, &c. the symbols of differentiation $\frac{d}{da}, \frac{d}{db}, \frac{d}{dc}$, &c.; in all that follows it is the successive evectants of the discriminant alone which come under consideration. I need hardly repeat, that the discriminant of a function is the result of the process of elimination (clear from extraneous factors) performed between the partial differential quotients of the function in respect to the several variables which it contains, or to speak more accurately, is the characteristic of their coevanescibility.

† The constitution of the quotients obtained by dividing all the other evectants of the discriminant by the first non-evanescent one, presents many remarkable features which remain yet to be fully studied out, and promise a wide extension of the existing theory.

as already remarked, this simply means that there are r distinct groups of equal roots, such groups containing $1+m_1$, $1+m_2 \ldots 1+m_r$ roots respectively. So for curves and higher loci, the total distributive multiplicity is the sum of the multiplicities at the several multiple points. But the true theory of the higher degrees of multiplicity separately considered at any point remains yet to be elaborated, and will be found to involve the consideration of the theory of elimination from a point of view under which it has never hitherto been contemplated.

Confining our attention for the present to curves, we have a clear notion of the multiplicity 1: this is what exists at an ordinary double point. As well known, its analytical character may be expressed by saying that the function of x, y, z, which characterizes the curve, is capable, when proper linear transformations are made, of being expanded under the form of a series descending according to the powers of z, such that the constant coefficient of the highest power of z, and the linear function of x, y, which is the coefficient of the next descending power of z, may both disappear. Again, when the multiplicity is 2, the third coefficient, which is a quadratic function of x and y, will become a perfect square. This is the case of a cusp, which, as I have said, is the precise analogue to that of three equal roots for a function of two variables. Before proceeding to consider what it is which constitutes a multiplicity 3 for a curve, it will be well to pause for a moment to fix the geometrical characters of the ordinary double point and the cusp.

If we agree to understand by a first polar to a curve the curve of one degree lower which passes through all the points in which the curve is met by tangents drawn from an arbitrary point taken anywhere in its own plane, we readily perceive that at an ordinary double point all the infinite number of first polars which can be drawn to the curve will intersect one another at the double point. Again, at a cusp all these polars will not only all intersect, they will moreover all touch one another at the cusp. Now we may proceed to inquire as to the meaning of a multiplicity of the third degree, which, strange to say, I believe has never yet been distinctly assigned by geometricians.

This is not the case of a so-called triple point, that is a point where three branches of the curve intersect. Supposing $x=0$, $y=0$, to represent such a point, the characteristic of the curve must be reducible to the form

$$(gx^3+hx^2y+kxy^2+ly^3)\,z^{n-3}+\&c.,$$

which, as is well known, involves the existence of four conditions. This, however, would not in itself be at all conclusive against the multiplicity at a triple point being only of the third degree; for it can readily be shown that there may exist singular points of any degree of *singularity* (as measured by the number of conditions necessary to be satisfied in order that such singularity may come into existence), but for which the multiplicity may be as low as we please; as, for instance, if at a double point (which is not a cusp) there be a point of inflexion on one branch or on both, or a point of undulation, or any other singularity whatever, still provided there be no cusps, the multiplicity will stick at the first degree and never exceed it; for only the discriminant itself will vanish on these suppositions, but no evectant of the discriminant. The reason, on the contrary, why a so-called triple point must be said to have a multiplicity of the degree 4, and not merely of the degree 3, springs from the fact that the 1st, 2nd and 3rd evectants of the discriminant all vanish at such a point.

It is clear, then, that there ought to exist a species of multiplicity for which the 1st and 2nd evectants vanish, but not the 3rd. In fact, as at a double point the first polars all merely intersect, but at a cusp have all a contact with one another of the first degree, so we ought to expect that there should exist a species of multiple point such that all the first polars should have with each other a contact of the second degree (or if we like so to say, the same curvature) at that point. When the curve has a triple point, all its first polars will have that point upon them as a double point; and it is not at the first glance, easy *à priori* to say what is the nature of the contact between two curves which intersect at a point which is a double point to each of them: we know upon settled analytical principles, that when one curve having a double point is crossed there by another curve not having a double point, that the two must be said to have with one another, a contact of the 1st degree; and we now learn from our theory of evection, that if each have a double point at the meeting-point, the degree of the contact must from principles of analogy be considered to be of the 3rd degree*. Now, then, we come to the question of deciding definitely what is a multiple point for which the degree of multiplicity is 3. It is, adopting either test, whether of first polar contact or of evection, a cusp situated or having its *nidus*, so to say, at a point of inflexion. In other words, $x=0$, $y=0$ will be a point whose multiplicity is intermediate between that of the cusp and that of a so-called triple point, when the characteristic of the curve admits of being written under the form

$$z^{n-2}x^2 + z^{n-3}(gx^3 + hx^2y + ixy^2) + z^{n-4} \text{ \&c.};$$

or in other words, when over and above the vanishing of the constant and linear coefficients, and the quadratic coefficient being a perfect square, as in the case of an ordinary cusp, this square has a factor in common with the next (the cubic) coefficient; or again, in other words, a curve has a point

* This may easily be verified by direct analytical means; as also the more general proposition, that two curves meeting at a point where there are m branches of the one and n branches of the other, must be considered to have mn coincident points in common, that is, if we like so to express it, to have a contact of the degree $mn-1$.

for which the multiplicity is 3 when its characteristic function admits of being expanded according to the powers of one of the variables, in such a manner that the first coefficient and the second (the linear) coefficient vanish, and that the discriminant of the third and the resultant of the third and fourth are both at the same time zero. This being the case, it may be shown that the first polars will all have with each other a contact of the second degree; and moreover, that all the evectants of the discriminant will have as a common factor a linear function of the variables, raised to a power whose index is three times that of the characteristic function. As, then, there is but one kind of ordinary double point, and but one kind of point with multiplicity 2, so there is one, and only one, kind of point with a multiplicity 3. A cusp is a peculiar double point; a flex-cusp (as for the moment I call the point last above discussed) is a peculiar cusp. This law of unambiguity, however, appears to stop at the third degree. A so-called triple point (which ought in fact to be called a *quintuple* point) is a point for which the multiplicity, as shown above, is of the fourth degree; but it is not the only point of that degree of multiplicity. Without assuming to have exhausted every possible supposition upon which such a degree of multiplicity may be brought into existence, it will be sufficient to take as an example a curve whose characteristic is capable of assuming the form

$$z^{n-2}x^2 + z^{n-3}(gx^3 + hx^2y) + z^{n-4}(kx^4 + lx^3y + mx^2y^2 + nxy^3) + z^{n-5}\,\&\text{c.}$$

It may readily be demonstrated that the first polars of this curve have all with one another at the point x, y a contact of a degree exceeding the 2nd, that is of at least the 3rd degree (and, I believe, in general not higher). Now the point x, y is evidently not a triple-branched point, but a cusp with three additional degrees of singularity; so that we have evidence of the existence of a point whose degree of singularity is 5, and whose multiplicity is at least 4, but which is in no sense a modified triple point. It is probably true (but to demonstrate this requires a further advance to be made than has yet been realized in the theory of the constitution of discriminants) that a cusp may be so modified by the *nidus* at which it is posited, as, without ever passing into a triple point, to be capable of furnishing any amount of multiplicity whatever, curiously in this contrasting with an ordinary double point, no amount whatever of extraordinary singularity imparted to which, or so to speak, to its *nidus*, can ever heighten its multiplicity so as to make it surpass the first degree without first converting it into a cusp. I may illustrate the nature of a flex-cusp by what happens to a curve of the third degree. When it breaks up into a conic and a right line, there are two ordinary double points; for the existence of these double points, as for the existence of a cusp, two conditions are required. When, however, the right line and conic touch one another (a *casus omissus* this in the works of the special geometers), the characters of the cusp and the point of inflexion are combined at the point of contact; the multiplicity is of the third degree, and the singularity also of a degree not exceeding this; three conditions only being necessary to be satisfied in order that a given cubic may degenerate into such a form; and it will be found that the discriminant and the first and second evectants thereof vanish for this case, and that the third evectant of the discriminant will be a perfect 9th power; whereas in order that the cubic may have a so-called triple point, that is may degenerate into a trident of diverging rays, four conditions must be satisfied, and it will be found that when this is the case, the first, second, and third evectants of the discriminant will all vanish, and the fourth will be a perfect 12th power of a linear function of the variables. I may mention, by the way, at this place, that the law of a discriminant and the successive evectants up to the mth inclusive, all vanishing, may be expressed otherwise (not in *identical*, but in *equivalent* or *equipollent* terms), by saying that the discriminant and all its derivatives of a degree not exceeding the mth will all vanish—understanding by a derivative of the discriminant any function obtained from the discriminant by differentiating it any specified number of times with respect to the constants of the function to which it belongs, the same constants being repeated or not indifferently*. And very surprising it must be allowed to be, stated as a bare analytical fact, that $(m + 1)$ conditions imposed upon the coefficients of a function of any number of variables and of any degree should suffice to make the inordinately greater number of functions which swarm among the derivatives of the mth and inferior degrees of the discriminant each and all simultaneously vanish.

Without pushing these observations too far for the patience of the general reader, it may be remarked by way of setting foot with our new theory upon the almost unvisited region of the singularities of surfaces, that by the light of analogy we may proceed with a safe and firm step as far as multiplicity of the third degree inclusive.

The function characteristic of the surface being supposed to be expressed in terms of the four variables x, y, z, t, and expanded according to descending powers of t, then when x, y, z is an ordinary double point of the first degree of multiplicity, the constant and the linear coefficient disappear; when the point has a multiplicity 2, the discriminant of the quadratic coefficient will be zero, that is this coefficient will be expressible by means of due linear transformations under the form of $x^2 + y^2$; and when the multiplicity is to be of the degree 3, the cubic coefficient will, at the same time that the quadratic coefficient is put under the form $x^2 + y^2$, itself (for the same system of x and y) assume the form of a cubic function of x, y, z, in which the highest power of z, that is z^3, will not appear; or in other words (restoring to x, y, z their

* Or, to speak more simply, the discriminant and its successive *differentials* up to the mth exclusive must all vanish simultaneously.

generality), not only will the first derivatives of the quadratic function be nullifiable simultaneously with each other, but likewise at the same time with the cubic function itself. These three cases will be for surfaces, the analogues so far, but only so far as regards the degree of the multiplicity, to the double point, cusp, and flex-cusp of curves*. The analogue to the so-called triple point of the curves will be a point whose degree of singularity, depending upon the vanishing of the six constants in the third coefficient (which is a quadratic function of x, y, z) at the same time as the three constants in the linear factor, would seem to be but 6 more than for a double point, that is in all $1+6$ or 7, but whose multiplicity, as inferred from the nature of the contact of its first polars, which will be of the 7th order, would appear to be 8 (a seeming incongruity which I am not at present in a condition to explain)†; so that there will apparently be 4 steps of multiplicity to interpolate between this case and the case analogous (*sub modo*) to the flex-cusp, last considered. Whether these intervening degrees correspond to singularities of an unambiguous kind, no one is at present in a condition to offer an opinion. I will conclude with a remark, the result of my experience in this kind of inquiry as far as I have yet gone in it, namely that it would be most erroneous to regard it as a branch of isolated and merely curious or fantastic speculation. Every singularity in a locus corresponds to the imposition of certain conditions upon the form of its characteristic; by aid of the theory of evection we are able to connect the existence of these conditions with certain consequences happening to the form of the discriminant, and thereby it becomes possible, upon known principles of analysis, to infer particulars relating to the constitution of the discriminant itself in its absolutely general form, very much upon the same principle as when the values of a function for particular values of its variable or variables are known, the general form of the function thereby itself, to some corresponding extent, becomes known. Thus, for instance, I have by the theory of evection in its most simple application, been led to a representation of the discriminant

* At an ordinary conical point of a surface for which the multiplicity is 1, every section of the surface is a curve with a double point. When the multiplicity is 2, the cone of contact becomes a pair of planes, through the intersection of which any other plane that can be drawn cuts the surface in a section having an ordinary cusp of multiplicity 2, but which themselves cut the surface in sections, having so-called triple points, so that for these two principal sections (which is rather surprising) the multiplicity suddenly jumps up from 2 to 4. All other things remaining unaltered when the multiplicity of the conical point is 3, the cusp belonging to any section of the surface drawn through any intersection of the two tangent planes passes from an ordinary cusp to a flex-cusp.

† So, too, at a so-called quadruple point in a curve, the degree of the contact of the 1st polars is 8, and therefore the multiplicity of the curve at such point is 9; but the number of constants which vanish for this case (namely all those of the cubic coefficient in x, y) over and above what vanish for the case of a so-called triple point is only 4, which is a unit less than the difference between the measures of the multiplicities at the respective points; and this difference continues to increase as we pass on to so-called quintuple and higher multiple points in the curves.

of a function of two variables under a form very different and very much more complete and fecund in consequences than has ever been supposed, or than I had myself previously imagined, to be possible.

According to the opinion expressed by an analyst of the French school, of pre-eminent force and sagacity, it is through this theory of multiplicity, here for the first time indicated, that we may hope to be able to bridge over for the purposes of the highest transcendental analysis, the immense chasm which at present separates our knowledge of the intimate constitution of functions of two from that of three, or any greater number of variables.

It is, as I take pleasure in repeating, to a hint from Mr Cayley*, who habitually discourses pearls and rubies, that I am indebted for the precious and pregnant observation on the form assumed by the first discriminantal evectant of a binary function with a pair of equal roots, out of which, combined with some antecedent reflections of my own, this new theory of multiplicity has taken its rise. The idea of the process of evection, and the discovery of its fundamental property of generating what, in my calculus of forms (*Cambridge and Dublin Mathematical Journal*), I have called *contravariants*, is due to my friend M. Hermite. The polar reciprocals of curves and other loci are contravariants and, as I have recently succeeded in showing, for curves at least, evectants, but of course not discriminantal evectants; and I am already able to give the actual explicit rule for the formation of the polar reciprocal of curves as high as the 5th degree, which with a little labour and consideration can be carried on to the 6th, and in fact to curves of any degree n when once we are acquainted with any mode of determining all such independent invariants of a function of two variables as are of dimensions not exceeding $2(n-1)$ in respect of the coefficients.

By the special geometers (by whom I mean those who, unvisited by a higher inspiration, continue to regard and to cultivate geometry as the science of mere sensible space) this problem has only been accomplished, and that but recently, for curves whose degrees do not exceed the 4th. Mr Salmon has made the happy and brilliant (and by the calculus of forms instantaneously demonstrable) discovery, communicated to me in the course of a most instructive and suggestive correspondence, that *a certain readily ascertainable*

* Mr Cayley's theorem stood thus:—If

$$ax^n + nbx^{n-1}y + \dots + nb'xy^{n-1} + a'y^n$$

have two equal roots, and ϖ be its discriminant, then will

$$\left\{y^n \frac{d}{da} - y^{n-1}x \frac{d}{db} \&\text{c.} \pm x^n \frac{d}{da'}\right\} \varpi$$

be a perfect nth power. It will easily be seen that this theorem is convertible into a theorem of evection by interchanging in the result x and y with y and $-x$.

*evectant of every discriminant of any function whatever is an exact power of its polar reciprocal**.

I believe that it may be shown, that, with the sole exception of odd-degreed functions of two variables, the *polar reciprocal itself* (as distinguished from a power thereof) of every function is an evectant, not (of course) of the discriminant, but of some determinable inferior invariant.

P.S. The terms pluri-simultaneous and pluri-simultaneity, used or suggested by me in my last paper in the *Magazine*, may be advantageously replaced by the more euphonious and regularly formed words consimultaneous, consimultaneity. Multiplicity and all its attributes and consequences are included as particular cases in the general conception and theory of consimultaneity, that is of consimultaneous equations, or, which is the same thing, of consimulevanescent functions.

* Namely, for a function of degree n, and variability (that is, having a number of variables) p, the $(n-1)^{p-1}$th evect of the discriminant is the $(n-1)$th power of the polar reciprocal.